PySpark installation steps for both Ubuntu and Amazon Linux.

Ubuntu Installation

Step 1: Update packages

sudo apt update

Step 2: Install Java

PySpark needs Java.

sudo apt install openjdk-11-jdk -y

Check Java:

java -version

Step 3: Install Python

sudo apt install python3 python3-pip -y

Check:

python3 --version
pip3 --version

Step 4: Install PySpark

pip3 install pyspark

Step 5: Verify PySpark

pyspark

You should see:

Welcome to Spark

Step 6: Exit

exit()

Optional environment variables:

nano ~/.bashrc

Add:

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export SPARK_HOME=/home/ubuntu/.local/lib/python3.10/site-packages/pyspark
export PATH=$PATH:$SPARK_HOME/bin

Reload:

source ~/.bashrc

Amazon Linux Installation

Step 1: Update server

sudo yum update -y

For Amazon Linux 2023:

sudo dnf update -y

Step 2: Install Java

sudo yum install java-11-amazon-corretto -y

Amazon Linux 2023:

sudo dnf install java-11-amazon-corretto -y

Check:

java -version

Step 3: Install Python and pip

sudo yum install python3 python3-pip -y

Amazon Linux 2023:

sudo dnf install python3 python3-pip -y

Check:

python3 --version

Step 4: Install PySpark

pip3 install pyspark

Step 5: Start PySpark

pyspark

Expected:

Spark session started

Use this command from Windows PowerShell or CMD.

Basic SCP command:

scp -i "C:\path\your-key.pem" "C:\path\file.txt" ec2-user@EC2_PUBLIC_IP:/home/ec2-user/

For Ubuntu EC2, username is usually:

ubuntu

Command:

scp -i "C:\path\your-key.pem" "C:\path\file.txt" ubuntu@EC2_PUBLIC_IP:/home/ubuntu/

For Amazon Linux EC2, username is usually:

ec2-user

Command:

scp -i "C:\path\your-key.pem" "C:\path\file.txt" ec2-user@EC2_PUBLIC_IP:/home/ec2-user/

Full steps:

Step 1: Keep your pem file and file ready

Example:

cd "C:\Users\Admin\Downloads"

icacls "C:\Users\Admin\Downloads\ spark.st.pem" /inheritance:r

icacls "C:\Users\Admin\Downloads\ spark.st.pem" /grant Admin:R

scp -i "C:\Users\Admin\Downloads\ spark.st.pem" "C:\Users\Admin\Downloads\customers-100000.csv" ec2-user@13.203.102.32:/home/ec2-user/

C:\Users\Admin\Downloads\europe.st.pem C:\Users\Admin\Downloads\customers-100000.csv

Step 2: Open PowerShell

Step 3: Run SCP command

For Amazon Linux:

scp -i "C:\Users\Yogesh\Downloads\ europe.st" " C:\Users\Admin\Downloads\customers-100000.csv " ec2-user@51.20.126.243:/home/ec2-user/

For Ubuntu:

scp -i "C:\Users\Yogesh\Downloads\my-key.pem" "C:\Users\Yogesh\Desktop\test.py" ubuntu@13.232.10.25:/home/ubuntu/

Step 4: Login to EC2 and check file

Amazon Linux:

ssh -i "C:\Users\Yogesh\Downloads\my-key.pem" ec2-user@13.232.10.25

Ubuntu:

ssh -i "C:\Users\Yogesh\Downloads\my-key.pem" ubuntu@13.232.10.25

for pem file WARNING: UNPROTECTED PRIVATE KEY FILE! Error

cd C:\Users\Admin\Downloads

icacls .\spark.pem /inheritance:r

icacls .\spark.pem /remove "DESKTOP-4L1ONKU\Codex"

icacls .\spark.pem /grant:r "$($env:USERNAME):(R)"

Check file:

ls -l

Copy folder from Windows to EC2:

scp -i "C:\Users\Yogesh\Downloads\my-key.pem" -r "C:\Users\Yogesh\Desktop\myfolder" ec2-user@1

sudo yum install java-17-amazon-corretto -y

sudo alternatives --config java

java -version

pyspark